

Kushagra Bhatnagar

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PROFESSIONAL SUMMARY

MSc student in Data Science and Machine Learning at the **National University of Singapore**, with an undergraduate degree in Economics from **IIT Kanpur**. Past work includes a **deployed data analysis agent**, a system that turns books into narrated lessons, a text to 3D scene generator, and **Python packages on PyPI** that run logic as explicit recorded steps, so any result can be replayed and checked. Recent experimental work tested when an AI coding agent can be trusted to judge that its own task is complete. Research output spans machine learning, signal processing, and trade economics, including an **IEEE best paper award**, a **Routledge book chapter**, and a preprint read **over 50,000 times**.

SYSTEMS

ADSP, a data analysis agent (deployed) · [demo](#)

An AI assistant built into a data analysis platform. It chooses and runs the right analysis steps for a request, can **carry out a complete analysis on its own**, scores its own results, and **logs every action** so the work can be reviewed afterwards.

RLang stack, published on PyPI · [rlang-compiler](#) · [semantic-compiler-core](#) · [bor-sdk](#)

Three Python packages that turn written rules into programs which run step by step, record each step, and let anyone **replay a run to verify the result**. **Provisional patent filed** at the Indian Patent Office, November 2025.

Tome, an AI tutor · [demo](#) · [code](#)

Upload a book and it produces **narrated slide lessons**. The book is split into small sections, and for each one the system pulls out the key points, arranges them on slides, and generates the matching narration, so a full book becomes a sequence of taught lessons.

Text to 3D

Turns a written description into a **rendered 3D scene**, by converting the text into an intermediate scene format that is then compiled into the final render.

RESEARCH AND PUBLICATIONS

1. [Who Decides When the Task Is Done?](#) Zenodo, 2026. Preregistered experiments, **2,862 episodes across seven models**, on whether a coding agent can be trusted to declare its own work complete. **Code and data public**.
2. **Gauntlet**, 2026. A benchmark of **452 code repair tasks** with machine-verified difficulty and solution properties, built for measuring how coding agents work through a task.
[GitHub Repository](#)
3. [Machine Learning Based Prediction of Wireless Fading Distribution Functions](#). IEEE World Conference on Applied Intelligence and Computing, 2025. **Best Paper Award**.
4. [Re-imagining ASEAN: Analysing the Trade Effects of India and Japan's Participation in the ASEAN Framework Using a Structural Gravity Model](#). **Book chapter, Routledge**, 2025.
5. [RAM](#) and [BoR](#), two whitepapers on making program execution **reproducible and verifiable**. Zenodo, 2025.
6. [Modeling and Predictions for COVID 19 Spread in India](#), 2020. **Over 50,000 reads**.

In progress: preregistered experiments on the Gauntlet corpus.

EXPERIENCE

CIFDAQ · Research Scientist

Jul 2025 to Nov 2025

Built a **trading signal framework** that models price movement as a **system of physical forces**. Authored research reports on blockchain activity, staking, and token economics that fed directly into product decisions.

Ernst & Young · Government Consulting Intern

May 2024 to Jul 2024

Reviewed how India and five other countries collect **balance of payments** data and proposed changes to align India's trade reporting with international practice.

DesignX · Data Analyst Intern

2023

Analyzed industrial production data using **ensemble learning methods** to find drivers of performance and operational bottlenecks.

TimesPro · Management Consulting Intern

2023

Competitive and pricing analysis for education sector products.

SKILLS

Programming: Python (NumPy, pandas), SQL, JavaScript, L^AT_EX

Working with LLMs: building agents with tool calling, retrieval augmented generation, evaluation harnesses

Engineering: git, testing, Docker, Kafka, packaging and publishing (three packages on PyPI)

Quantitative methods: statistics, econometrics, causal inference, game theory

EDUCATION

National University of Singapore · MSc, Data Science and Machine Learning

Aug 2026 to Dec 2027 (expected)

IIT Kanpur · BS, Economics · **CPI 8.8 out of 10**

2021 to 2025

Department rank one in first year, KN Saluja Scholarship, Academic Excellence Awards in 2021/22 and 2022/23.